

32-bit Memory Interfacing (Intel 80486)

A hardware design project implementing a **1 MB SRAM memory interface** for the Intel 80486 microprocessor, developed at **Nile University**.

Team Members

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Project Overview

This project designs and implements a complete **1 MB Static RAM (SRAM) memory subsystem** for the Intel 80486 processor. The system features a 32-bit data bus organized into four independent 8-bit memory banks, mapped to the base address **12000000H**.

Hardware Specifications

Parameter	Value
Processor	Intel 80486
Data Bus Width	32-bit
Total Memory	1 MB (1,024 KB)
SRAM Chip	62512 (64 KB each)
Total Chips	16 (4 rows × 4 banks)

Start Address	12000000H
End Address	120FFFFFFH
Bank Organization	4 banks × 8 bits



Address Mapping

Global Address Range

Start: 12000000H
End: 120FFFFFFH
Size: 100000H (1 MB)

Address Bus Allocation (A31–A0)

Bits	Lines	Function
A31–A20	12 lines	Base address decoding (fixed = 120H)
A19–A18	2 lines	Row selection (input to 74138 decoder)
A17–A2	16 lines	Chip internal offset (→ SRAM A0–A15)
A1–A0	—	Not used; replaced by BE0#–BE3# signals

Row Address Map

Row	Start Address	End Address	A19	A18	Decoder Output
0	12000000H	1203FFFFFFH	0	0	Y0
1	12040000H	1207FFFFFFH	0	1	Y1
2	12080000H	120BFFFFFFH	1	0	Y2
3	120C0000H	120FFFFFFH	1	1	Y3

Logic Design

1. Master Chip Select (Master_CS#)

Decoded from the upper 12 address bits using a **12-input NAND gate** with inverters on lines that must be logic '0':

$$\text{Master_CS\#} = \text{NAND}(\text{A28}, \text{A25}, \bar{\text{A31}}, \bar{\text{A30}}, \bar{\text{A29}}, \bar{\text{A27}}, \bar{\text{A26}}, \bar{\text{A24}}, \bar{\text{A23}}, \bar{\text{A22}}, \bar{\text{A21}}, \bar{\text{A20}})$$

2. Row Selection — 74138 3:8 Decoder

Pin	Connection
Input A	Processor A18
Input B	Processor A19
Input C	GND (Logic 0)
G1 (Active High)	M/IO#
G2A# (Active Low)	ADS#
G2B# (Active Low)	Master_CS#
Outputs Y0–Y3	CS# of each memory row

3. Bank Write Control (OR Gates)

Byte-level write control combining MWTC# with individual byte enables:

$$\begin{aligned}\text{WR0\#} &= \text{MWTC\#} \text{ OR } \text{BE0\#} && (\text{Bank 0} - \text{D0-D7}) \\ \text{WR1\#} &= \text{MWTC\#} \text{ OR } \text{BE1\#} && (\text{Bank 1} - \text{D8-D15}) \\ \text{WR2\#} &= \text{MWTC\#} \text{ OR } \text{BE2\#} && (\text{Bank 2} - \text{D16-D23}) \\ \text{WR3\#} &= \text{MWTC\#} \text{ OR } \text{BE3\#} && (\text{Bank 3} - \text{D24-D31})\end{aligned}$$

Interconnection Map

Processor Signal	Connected To	Description
A2–A17	A0–A15 (SRAM)	Internal chip addressing

A18, A19	A, B (74138)	Row selection
A20–A31	NAND Logic	Master_CS# generation
D0–D7	Bank 0	Least significant byte
D8–D15	Bank 1	—
D16–D23	Bank 2	—
D24–D31	Bank 3	Most significant byte
MRDC#	OE# (All chips)	Read enable
WR(n)#	WE# (Bank n)	Bank-specific write enable
M/IO#	G1 (74138)	Memory cycle qualifier
ADS#	G2A# (74138)	Address valid signal

Design Highlights

-  Supports 8-bit, 16-bit, and 32-bit data transfers via byte-enable logic
-  Assumes SRAM access time (15–20 ns) compatible with 80486 bus clock (no wait states needed)
-  Expandable: 74138 outputs Y4–Y7 available for an additional 1 MB

File Structure

```
memory-interfacing-80486/
|
├─ schematic/
|   └─ memory_schematic.jpg      # Hand-drawn architectural schematic
├─ report/
|   └─ PROJECT_MICRO.pdf        # Full project report
└─ README.md
```

References

- Intel. *Intel 80486 Microprocessor Hardware Reference Manual*
- Texas Instruments. *SN74LS138 Decoder/Demultiplexer Datasheet*
- ISSI. *IS62C512 — 512K (64K × 8) SRAM Datasheet*